Lecture 1

Lie groups and their tangent algebras

Objectives. *A Lie group has both a group structure and a smooth geometry. We describe that geometry near the identity and recover its Lie algebra. The group* $\mathrm{SU}(2)$ *will give an example in which every step can be calculated.*

By the end of this lecture you will be able to:

- Identify the real dimension of a matrix Lie group.
- Find tangent spaces by differentiating a defining condition and exhibiting curves.
- Recover a matrix Lie bracket from left translations.
- Distinguish a real Lie algebra from its complexification.

The Lie algebras in the earlier course were complex. The tangent algebra of a real Lie group is a real Lie algebra. We will always state the field of scalars.

1.1 Smooth coordinates

A sphere cannot be described by one set of Euclidean coordinates without omitting a point or introducing a singularity. It can be described by several overlapping coordinate systems. A smooth manifold formalises this local description.

A topology specifies which subsets are open. A *neighbourhood* of a point is a set containing an open set around that point. A topological space is *Hausdorff* if distinct points have disjoint open neighbourhoods. It is *second countable* if its topology has a countable *basis*, namely a countable collection of open sets such that every open set is a union of some of them.

A map between topological spaces is *continuous* if the inverse image of every open set is open. A *homeomorphism* is a continuous bijection whose inverse is continuous.

Definition 1.1 (Smooth manifold)**.** A smooth real manifold of dimension d is a topological space M covered by coordinate charts

$$\varphi : U \longrightarrow V \subseteq \mathbb{R}^d.$$

Each chart has open domain $U \subseteq M$ and is a homeomorphism onto an open set V. Where the domains of two charts φ and ψ overlap, the coordinate change $\psi \circ \varphi^{-1}$ is infinitely differentiable. We require M to be Hausdorff and second countable.

A subset of a topological space has the *subspace topology*: its open sets are its intersections with open sets of the space. For subsets of Euclidean space we use this topology. For example, an open arc is open in the circle.

Subsets of Euclidean space are Hausdorff and second countable, so all the matrix groups in this lecture satisfy both conditions. Lecture 2 uses the Hausdorff condition for uniqueness of solutions of differential equations on a manifold. Lecture 3 uses second countability for homogeneous spaces. A smooth map between manifolds is a continuous map whose coordinate expressions are infinitely differentiable on their open domains. A *diffeomorphism* is a smooth bijection whose inverse is smooth.

Example 1.2 (A coordinate on the circle)**.** The unit circle is

$$S^1 = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = 1\}.$$

On the open right semicircle, use y as the coordinate. Its inverse chart is

$$y \longmapsto \left(\sqrt{1 - y^2}, y\right), \qquad -1 < y < 1.$$

The other three semicircles have similar charts. On the upper semicircle use x. On the open quarter circle where both charts apply, y runs over $(0, 1)$, and the coordinate change and its inverse are

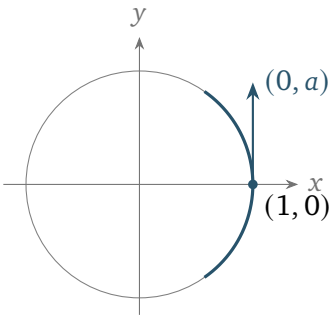
$$x = \sqrt{1 - y^2}, \qquad y = \sqrt{1 - x^2}.$$

These are smooth because $1 - y^2 > 0$ and $1 - x^2 > 0$ there. The other overlaps are similar. Thus the circle is a one-dimensional real manifold, although it lies in a two-dimensional ambient space.

The coordinate y can also describe a velocity. For a real number a, the curve on the circle

$$\gamma_a(t) = \left(\sqrt{1 - a^2 t^2}, at\right)$$

is defined near zero. It starts at $(1, 0)$ and has ambient derivative $(0, a)$. The derivative of its y-coordinate is a. Thus a one-dimensional tangent space can appear as a line inside a larger ambient space.



The coloured arc lies in the domain of one chart, in which y is the local coordinate. The tangent arrow is the velocity at $(1, 0)$ for positive a. Its drawn length is schematic.

1.2 Smooth groups

Product charts $\varphi \times \psi$ on $U \times V$ give $M \times N$ a smooth manifold structure of dimension $\dim M + \dim N$. A finite product of Hausdorff, second-countable spaces has both properties.

Definition 1.3 (Lie group)**.** A real Lie group is a group G that is also a smooth real manifold, with smooth multiplication and inversion:

$$G \times G \longrightarrow G, \quad (g,h) \longmapsto gh, \qquad G \longrightarrow G, \quad g \longmapsto g^{-1}.$$

A Lie group homomorphism is a smooth group homomorphism.

The additive group $\mathbb{R}^d$ is a Lie group. Any finite group with the discrete topology is also a Lie group, since each one-point subset is the domain of a chart modelled on $\mathbb{R}^0$. The tangent space of a finite group therefore has dimension zero. Hence a Lie algebra cannot distinguish finite groups.

Example 1.4 (General linear groups)**.** The group of invertible real matrices is

$$\mathrm{GL}_n(\mathbb{R}) = \{g \in M_{n\times n}(\mathbb{R}) : \det g \neq 0\}.$$

The determinant is continuous, so $\mathrm{GL}_n(\mathbb{R})$ is open in $\mathbb{R}^{n^2}$. Hence its dimension is n^2. The entries of a matrix product are polynomials in the entries of the factors, so matrix multiplication is smooth. Inversion is smooth because

$$g^{-1} = \frac{\operatorname{adj}(g)}{\det g}.$$

The same argument applies to $\mathrm{GL}_n(\mathbb{C})$. The real and imaginary parts of the entries give its coordinates, so its dimension as a real manifold is $2n^2$.

A *complex Lie group* has complex charts and holomorphic group operations. The group $\mathrm{GL}_n(\mathbb{C})$ has this extra structure. Its complex dimension is n^2. A group of complex matrices need not be a complex Lie group. Its defining conditions may involve complex conjugation.

For example, the unitary and special unitary groups are

$$\mathrm{U}(n) = \{U \in M_{n\times n}(\mathbb{C}) : U^*U = I\}, \qquad \mathrm{SU}(n) = \{U \in \mathrm{U}(n) : \det U = 1\}.$$

Here U^* denotes conjugate transpose. Their real Lie-group structures are a standard classical-group result, cited from Bröcker–tom Dieck, Chapter I, Section 1. We verify the manifold structure of $\mathrm{SU}(2)$ directly; Exercise Sheet 1 treats $\mathrm{U}(n)$.

A manifold is *connected* if it cannot be written as the union of two nonempty disjoint open subsets. For example, the orthogonal group is

$$\mathrm{O}(n) = \{g \in M_{n\times n}(\mathbb{R}) : g^{\mathsf{T}}g = I\}, \qquad n \geq 1.$$

Its determinant takes values in $\{1,-1\}$. Both values occur. The identity has determinant one, and $\operatorname{diag}(-1,1,\ldots,1)$ has determinant minus one. The inverse images of these two values are disjoint nonempty open subsets that cover the group. Thus $\mathrm{O}(n)$ is disconnected. Its subgroup with determinant one is $\mathrm{SO}(n)$.

1.3 Tangent vectors at the identity

Definition 1.5 (Tangent vector). Let $\varphi : U \to V$ be a chart of M with $p \in U$. A *smooth curve through* p is a smooth map γ from an open interval containing 0 to M with $\gamma(0) = p$. Two such curves γ and δ are *equivalent* when $(\varphi \circ \gamma)'(0) = (\varphi \circ \delta)'(0)$. A *tangent vector* at p is an equivalence class $[\gamma]$. The set of tangent vectors at p is denoted T_pM. The map $[\gamma] \mapsto (\varphi \circ \gamma)'(0)$ identifies T_pM with $\mathbb{R}^d$. Vector addition and real scalar multiplication on T_pM are transferred through this identification.

The class $[\gamma]$ is the *velocity* of γ, also written $\gamma'(0)$. The chain rule makes the relation between curves independent of the chart. The derivative of a coordinate change is linear, so the vector-space structure on T_pM is also independent of the chart.

A smooth map $f : M \to N$ sends tangent vectors to tangent vectors. If v is represented by γ, the image of v is the velocity of $f \circ \gamma$. This defines the *differential*

$$df_p : T_pM \longrightarrow T_{f(p)}N, \qquad df_p\big(\gamma'(0)\big) = (f \circ \gamma)'(0).$$

In charts this is the usual derivative, so it is real linear. The chain rule gives $d(f \circ h)_p = df_{h(p)} \circ dh_p$.

Our groups lie inside matrix spaces, and we calculate with matrix entries. An *embedded submanifold* is the notion that justifies this. It has the subspace topology. Near each of its points, a smooth change of ambient coordinates turns it into a coordinate subspace. For example, near $(1, 0)$ the coordinates $(r-1, \theta)$ turn the circle into the line $r-1 = 0$. We import the standard properties of an embedded submanifold. Its inclusion is smooth with injective differential. Products of embedded submanifolds are embedded in the corresponding product. Thus smooth ambient maps restrict smoothly to these submanifolds. A smooth ambient map whose image lies in an embedded submanifold is smooth as a map into that submanifold. The differential of the inclusion identifies its tangent vectors with ambient velocities.

We also use one result from multivariable calculus. Let $U \subseteq \mathbb{R}^N$ be open, and suppose a smooth map $F : U \to \mathbb{R}^r$ has surjective differential at every point of $F^{-1}(c)$. The *regular level-set theorem* says that $F^{-1}(c)$ is an embedded submanifold of U of dimension $N - r$, with the subspace topology. The tangent space of $F^{-1}(c)$ at a point p is $\ker dF_p$. The regular level-set theorem and the embedded-submanifold properties are cited from Warner, Chapter 1, sections “Submanifolds, Diffeomorphisms, and the Inverse Function Theorem” and “Implicit Function Theorems” (Theorem 1.38). For example, the circle is $F^{-1}(1)$ with $F(x, y) = x^2 + y^2$. Here $dF_{(x,y)} = (2x, 2y)$ is nonzero on the circle, so the dimension is $2 - 1 = 1$.

The kernel clause of the regular level-set theorem now follows from the chain rule and rank–nullity. Put $P = F^{-1}(c)$ and let $i : P \to U$ be the inclusion. For $p \in P$, since $F \circ i$ is constant, $dF_p \circ di_p = 0$. Thus $di_p(T_pP) \subseteq \ker dF_p$. The inclusion is an embedding, so di_p is injective. Both spaces have dimension $N - r$, because dF_p is surjective. Hence they are equal.

For an embedded manifold in a matrix space, we compute velocities by differentiating the matrix entries of curves. The identity element of a group is denoted e, or I for a matrix group. For a Lie group G, the vector space of interest is

$$\mathfrak{g} = T_eG.$$

For $\mathrm{GL}_n(\mathbb{R})$, every matrix X is a tangent vector at I. Indeed, the curve $\gamma(t) = I+tX$ is invertible for sufficiently small t, and $\gamma'(0) = X$.

Example 1.6 (Tangent vectors to the orthogonal group)**.** Let $g(t)$ be a smooth curve in $\mathrm{O}(n)$ with $g(0) = I$. Put $X = g'(0)$. Differentiate the defining identity at zero by the product rule:

$$g(t)^{\mathsf{T}} g(t) = I \quad \Longrightarrow \quad X^{\mathsf{T}} + X = 0.$$

Thus every tangent vector at I is skew-symmetric. Differentiating a condition gives a necessary condition. We must also show that every such matrix occurs.

We could use the exponential series, which Lecture 2 develops. The Cayley curve needs no convergence argument. For a skew-symmetric matrix X, consider the Cayley curve

$$g_X(t) = \left(I + \frac{tX}{2}\right)\left(I - \frac{tX}{2}\right)^{-1}.$$

It is defined near zero. Set $A = I+tX/2$ and $B = I-tX/2$. Then $A^{\mathsf{T}} = B$, $B^{\mathsf{T}} = A$, and $AB = BA$. Therefore

$$g_X(t)^{\mathsf{T}} g_X(t) = A^{-1}BAB^{-1} = I.$$

The determinant is continuous, takes values in $\{1, -1\}$, and equals one at zero. It remains one near zero. Hence this curve lies in $\mathrm{SO}(n)$. The product rule gives

$$g_X'(0) = \frac{X}{2} + \frac{X}{2} = X.$$

Consequently both orthogonal groups have the same tangent space at I:

$$T_I \,\mathrm{SO}(n) = T_I \,\mathrm{O}(n) = \{X : X^{\mathsf{T}} = -X\}.$$

The example used that $\mathrm{O}(n)$ is a smooth manifold. To justify this assertion, apply the regular level-set theorem to $F(g) = g^{\mathsf{T}}g$, with target the real symmetric matrices. At an orthogonal matrix g,

$$dF_g(H) = H^{\mathsf{T}}g + g^{\mathsf{T}}H.$$

For any symmetric matrix S, choose $H = gS/2$. Since S is symmetric and g is orthogonal, $dF_g(H) = S$. Hence the surjectivity hypothesis holds. Subtracting the dimension of the space of real symmetric matrices from the ambient dimension gives

$$\dim_{\mathbb{R}} \mathrm{O}(n) = n^2 - \frac{n(n+1)}{2} = \frac{n(n-1)}{2}.$$

The kernel clause gives the tangent space again: $dF_I(H) = H^{\mathsf{T}} + H$, so $T_I \,\mathrm{O}(n)$ consists of the skew-symmetric matrices. The Cayley curve showed the same thing by hand. The subgroup $\mathrm{SO}(n)$ is open in $\mathrm{O}(n)$, so it has the same local dimension.

Multiplication and inversion by transposition are restrictions of smooth matrix maps. By the embedded-submanifold properties, both orthogonal groups are Lie groups. Equal tangent spaces do not imply equal groups.

1.4 Translations and the bracket

A tangent space at one point is only a vector space, and it has no product. Vector fields do have a commutator, because they act on functions by differentiation. Left translation carries a tangent vector at e to a vector field on G. We therefore obtain a bracket on the tangent space. Lecture 2 shows that this bracket measures how far small group elements fail to commute.

Left multiplication by $g \in G$ is the smooth map

$$L_g : G \longrightarrow G, \qquad L_g(h) = gh.$$

Since L_g has inverse $L_{g^{-1}}$, the differential of L_g at e identifies tangent spaces:

$$(dL_g)_e : T_eG \longrightarrow T_gG.$$

For a matrix group this differential sends X to gX. For example, in $\mathrm{SO}(2)$ take the rotation and the tangent matrix

$$g = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}, \qquad X = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

Left translation gives the tangent vector at g:

$$gX = \begin{pmatrix} -\sin\theta & -\cos\theta \\ \cos\theta & -\sin\theta \end{pmatrix}.$$

This is the derivative of the rotation matrix with angle $\theta + t$ at $t = 0$.

A *vector field* assigns a tangent vector to each point. It is smooth if its components in each coordinate chart are smooth functions. The field determined by $X \in T_eG$ is

$$X^L(g) = (dL_g)_e X.$$

A vector field V on G is *left-invariant* if

$$(dL_h)_g\big(V(g)\big) = V(hg) \qquad \text{for all } g, h \in G.$$

The field X^L is left-invariant. Indeed, $L_h \circ L_g = L_{hg}$, so the chain rule gives $(dL_h)_g\big((dL_g)_e X\big) = (dL_{hg})_e X$.

Smooth vector fields act on smooth functions by directional differentiation. For a smooth vector field V and a smooth function f,

$$(Vf)(p) = df_p(V(p)).$$

Their commutator as differentiation operators is

$$[V, W]f = V(Wf) - W(Vf).$$

Theorem 1.7 (The tangent Lie algebra; cited). *The fields X^L obtained by left translation are smooth. The commutator of two smooth vector fields is again a vector field. The commutator of left-invariant fields is left-invariant. The value of this commutator at the identity defines a real Lie bracket on T_eG. For smooth vector fields V and W, in coordinates,*

$$[V, W](p) = DW_p(V(p)) - DV_p(W(p)).$$

For an embedded submanifold, this formula may be evaluated using smooth ambient extensions of tangent fields. The result is tangent to the submanifold.

The construction of the bracket from left-invariant fields is cited from Bröcker–tom Dieck, Chapter I, Section 2. The coordinate formula and the statement for embedded submanifolds are cited from Warner, Chapter 1, sections “Vector Fields” and “Submanifolds, Diffeomorphisms, and the Inverse Function Theorem”. The Jacobi identity is the commutator identity for the corresponding differentiation operators.

For an embedded matrix group, use the ambient extensions $X^L(g) = gX$ and $Y^L(g) = gY$. Since $Y^L(h) = hY$ is linear in h, its derivative in direction H is HY. Thus ambient calculus gives

$$D(Y^L)_g(gX) = gXY,$$
$$D(X^L)_g(gY) = gYX.$$

By the coordinate formula in the cited theorem, the vector-field bracket is the first of these derivatives minus the second. At the identity, this gives

$$[X, Y] = XY - YX.$$

This is the bracket studied in the earlier course. The cited submanifold compatibility shows that this ambient calculation is the intrinsic bracket on G. Hence $XY - YX$ belongs to T_IG. For skew-symmetric matrices, closure can also be seen directly:

$$(XY - YX)^{\mathsf{T}} = YX - XY = -(XY - YX).$$

The real tangent Lie algebra of $\mathrm{SO}(n)$ is denoted $\mathfrak{so}(n)$. For $\mathrm{GL}_n(\mathbb{C})$, viewed as a real group, the tangent algebra is all complex matrices with real scalars and the same commutator.

1.5 The group SU(2) and its real algebra

The two rows of a matrix in $\mathrm{SU}(2)$ are orthonormal. Write the first row as (α, β). The unit vectors orthogonal to this row are the multiples of $(-\overline{\beta}, \overline{\alpha})$ by complex numbers of modulus one. The determinant then equals that multiple, so determinant one forces it to be one. Hence every element of $\mathrm{SU}(2)$ has the form

$$U = \begin{pmatrix} \alpha & \beta \\ -\overline{\beta} & \overline{\alpha} \end{pmatrix}, \qquad |\alpha|^2 + |\beta|^2 = 1.$$

Conversely, a matrix of the displayed form satisfies

$$\begin{pmatrix} \overline{\alpha} & -\beta \\ \overline{\beta} & \alpha \end{pmatrix} \begin{pmatrix} \alpha & \beta \\ -\overline{\beta} & \overline{\alpha} \end{pmatrix} = \begin{pmatrix} |\alpha|^2 + |\beta|^2 & 0 \\ 0 & |\alpha|^2 + |\beta|^2 \end{pmatrix}.$$

The matrix U has determinant $|\alpha|^2 + |\beta|^2$. Thus this formula identifies $\mathrm{SU}(2)$ with the unit sphere S^3 in $\mathbb{C}^2 = \mathbb{R}^4$. The regular level-set theorem gives real dimension three. The displayed map from (α, β) to U is a real-linear injection from $\mathbb{R}^4$ into $M_{2\times 2}(\mathbb{C})$. It is a linear coordinate isomorphism onto its image, so it carries the embedded sphere to an embedded submanifold of the full matrix space. Multiplication and inversion are smooth restrictions of the matrix operations.

Proposition 1.8 (The tangent algebra of SU(2)). *The tangent Lie algebra of* $\mathrm{SU}(2)$ *is the real space*

$$\mathfrak{su}(2) = \left\{ \begin{pmatrix} ix & y + iz \\ -y + iz & -ix \end{pmatrix} : x, y, z \in \mathbb{R} \right\},$$

with the matrix commutator.

Proof. First we prove the necessary conditions. Let $U(t)$ be a smooth curve in $\mathrm{SU}(2)$ with $U(0) = I$, and put $X = U'(0)$. Differentiating $U(t)^* U(t) = I$ at zero gives $X^* = -X$.

The determinant condition requires a separate derivative. Write $X = (x_{jk})$. To first order in t,

$$U(t) = \begin{pmatrix} 1 + tx_{11} + o(t) & tx_{12} + o(t) \\ tx_{21} + o(t) & 1 + tx_{22} + o(t) \end{pmatrix}.$$

Its determinant has expansion

$$\det U(t) = 1 + t(x_{11} + x_{22}) + o(t).$$

Here $o(t)$ denotes a quantity whose ratio to t tends to zero. The products of two off-diagonal entries have order t^2, so they do not contribute to the derivative. Differentiating $\det U(t) = 1$ now gives $\operatorname{tr} X = 0$.

A skew-Hermitian matrix has purely imaginary diagonal entries. Its entry below the diagonal is minus the conjugate of the entry above it. Since the trace is zero, the diagonal entries are ix and $-ix$ for some $x \in \mathbb{R}$. Writing the entry above the diagonal as $y + iz$ gives the displayed form.

Next we prove the converse. Every matrix of the displayed form is a real linear combination of

$$A = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}, \qquad B = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad C = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}.$$

Let $X = xA + yB + zC$ and $r^2 = x^2 + y^2 + z^2$. Direct multiplication gives

$$X^2 = -r^2 I, \qquad \det(I + tX) = 1 + t^2 r^2.$$

These identities provide a curve with the required velocity:

$$U_X(t) = \frac{I + tX}{\sqrt{1 + t^2 r^2}}.$$

The scalar factor appears squared in a 2×2 determinant. Hence the determinant of $U_X(t)$ is one. Since $X^* = -X$ and $X^2 = -r^2 I$,

$$U_X(t)^* U_X(t) = \frac{(I - tX)(I + tX)}{1 + t^2 r^2} = I.$$

Also $U_X(0) = I$ and $U_X'(0) = X$. Hence every matrix of the displayed form is tangent to $\mathrm{SU}(2)$ at I.

Finally, $\mathrm{SU}(2)$ is an embedded matrix group. By the calculation with ambient extensions in the previous section, its bracket is the matrix commutator. ∎

Each group has so far needed its own curve. Lecture 2 gives a single one, $\exp(tX)$.

The multiplication of the basis matrices starts with $AB = C$ and $BA = -C$. The next products confirm a cyclic pattern:

$$BC = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix} = A, \qquad CB = \begin{pmatrix} -i & 0 \\ 0 & i \end{pmatrix} = -A.$$

Similarly, $CA = B$ and $AC = -B$. The bracket table is

$$[A, B] = 2C, \qquad [B, C] = 2A, \qquad [C, A] = 2B.$$

The scalars in this basis expansion are real. Multiplying A by i gives a Hermitian matrix, which is not in $\mathfrak{su}(2)$. The fact that its elements have complex entries does not make $\mathfrak{su}(2)$ a complex vector space. In physics, with the Pauli matrices $\sigma_1, \sigma_2, \sigma_3$ of *Linear Algebra for Physics*, one has $A = i\sigma_3$, $B = i\sigma_2$ and $C = i\sigma_1$.

The weight calculations of the earlier course use the matrices e and f given below. These matrices are not skew-Hermitian, so they are not in $\mathfrak{su}(2)$. To use them we must allow complex combinations of the real generators. We now relate this real algebra to the earlier complex algebra $\mathfrak{sl}_2(\mathbb{C})$.

Definition 1.9 (Complexification)**.** The complexification of a real Lie algebra $\mathfrak{k}$ is the complex vector space

$$\mathfrak{k}_{\mathbb{C}} = \mathfrak{k} \otimes_{\mathbb{R}} \mathbb{C},$$

with the complex-bilinear extension of its bracket. Thus $[X \otimes a, Y \otimes b] = [X, Y] \otimes ab$.

Let $\iota\colon \mathfrak{su}(2)_{\mathbb{C}} \to \mathfrak{sl}_2(\mathbb{C})$ be the complex-linear map given on pure tensors by $\iota(X \otimes a) = aX$. It extends the inclusion of $\mathfrak{su}(2)$ into complex matrices. Its values are traceless because the elements of $\mathfrak{su}(2)$ are traceless. The images of A, B, C span all traceless complex matrices, since

$$h = -iA, \qquad e = \frac{B - iC}{2}, \qquad f = -\frac{B + iC}{2}$$

give the three matrices

$$h = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad e = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad f = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}.$$

They are the usual basis matrices of $\mathfrak{sl}_2(\mathbb{C})$. These three matrices are complex linearly independent. The map ι preserves brackets by complex bilinearity. Its domain and image both have complex dimension three, so ι is a complex Lie algebra isomorphism:

$$\mathfrak{su}(2)_{\mathbb{C}} \cong \mathfrak{sl}_2(\mathbb{C}).$$

The real algebra has dimension three over $\mathbb{R}$. Its complexification has dimension three over $\mathbb{C}$. The real algebra underlying $\mathfrak{sl}_2(\mathbb{C})$ has dimension six over $\mathbb{R}$; it is a different object.

Self-checks

1. Why is $\mathrm{GL}_n(\mathbb{C})$ open in $\mathbb{C}^{n^2}$ but $\mathrm{U}(n)$ is not?
2. Does differentiating $g(t)^{\mathsf{T}} g(t) = I$ prove that every skew-symmetric matrix is tangent to $\mathrm{SO}(n)$?
3. Is $\mathrm{diag}(1, -1)$ in $\mathfrak{su}(2)$? Is it in the complex span of $\mathfrak{su}(2)$?

Answers. (1) The determinant is continuous, so the set where it is nonzero is open in $\mathbb{C}^{n^2}$. For real ε with $0 < |\varepsilon| < 1$, a unitary matrix U satisfies

$$((1+\varepsilon)U)^*((1+\varepsilon)U) = (1+\varepsilon)^2 I \neq I.$$

These non-unitary matrices approach U, so no ambient ball about U lies in $\mathrm{U}(n)$. (2) No. Differentiation proves necessity. The Cayley curve proves sufficiency. (3) No. The matrix $\mathrm{diag}(1, -1)$ is traceless but Hermitian, so it is not skew-Hermitian. It does lie in the complex span, since it equals $-iA$.

Summary and reading

A Lie group has smooth multiplication and inversion. Its tangent space at the identity is a real Lie algebra. Left translation determines the bracket. For embedded matrix groups, the bracket is the matrix commutator. The algebra records local information. Distinct groups can have the same tangent algebra.

Read Hall, second edition, Sections 1.1–1.5 and 3.3–3.6, and Bröcker–tom Dieck, Chapter I, Sections 1–2. The regular level-set theorem and the embedded-submanifold properties are the geometric imports, cited from Warner, Chapter 1. Lecture 2 develops the exponential map and connects these tangent vectors to one-parameter subgroups.